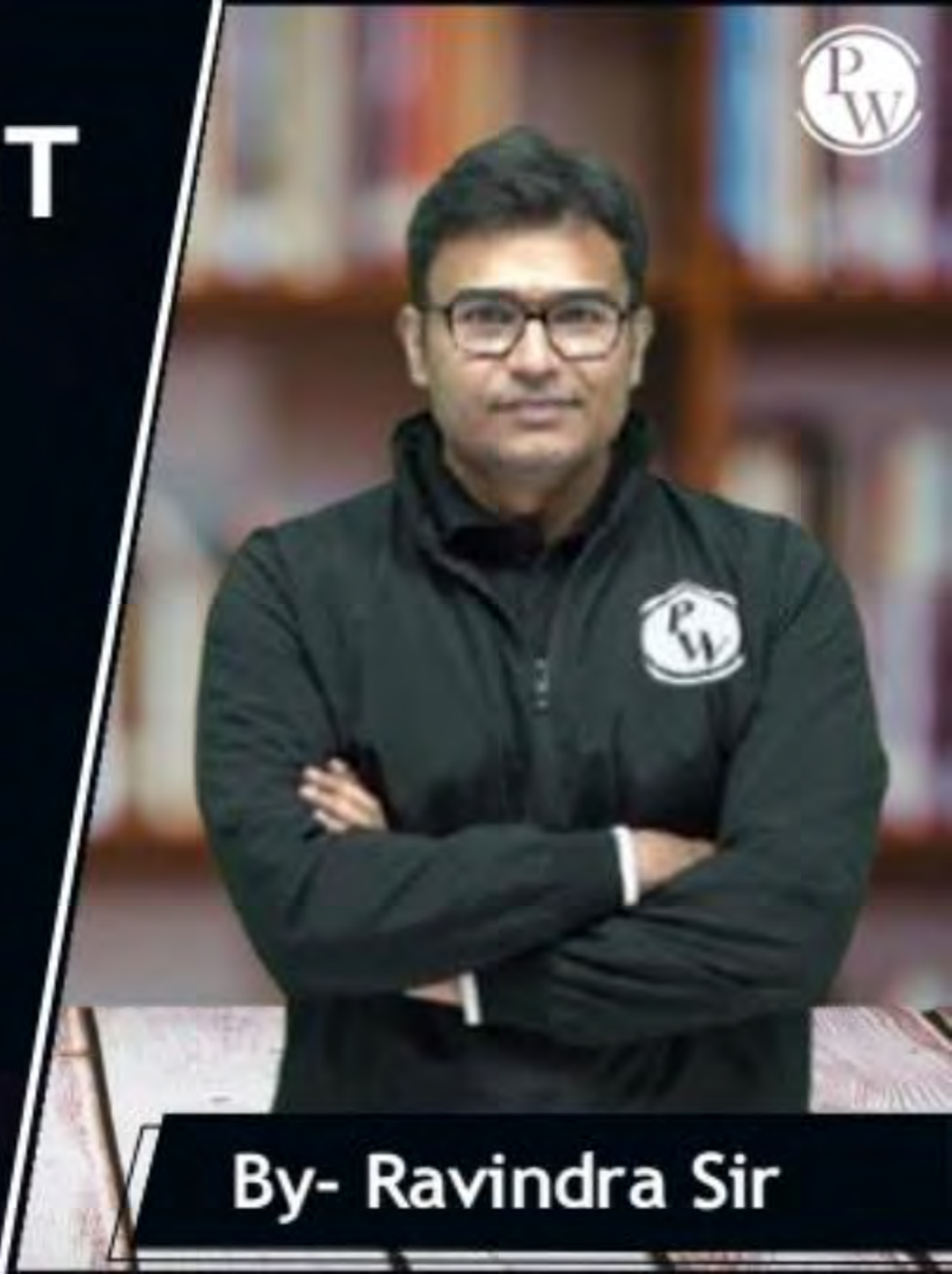
Computer Science & IT

COMPUTER NETWORKS (CN)

Flow Control Methods

Lecture No. 05



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Flow Control
CSMA/CD

$T_t \geq 2 * T_p$

$L \geq 2 \times T_p \times BW$

Topics to be Covered



- Topic
- Topic
- Topic

Flow Control & EC

↓

CSMA/CD



Inspiring Stories : Pulin Behari Das

Pulin Behari Das Background: Born 1877 in Bengal; a quiet-looking man who ran secret gyms that trained fighters.

Struggle: British spies hunted these clubs. Cops raided them, members were jailed or betrayed.

Education: Street-smart, organised member of the radical Anushilan Samiti.

Achievement: Built secret bomb workshops, trained youth in fighting and stealth, and ran an underground network that pulled off robberies and raids to fund revolution. He was the brains behind a shadow army.

Impact: Turned ordinary gym halls into a city-wide secret fighting force that the British feared.



Inspiring Stories : Bagha Jatin

Bagha Jatin (Jatindranath Mukherjee) Background: Big, tough, calm — called "Bagha" after he reportedly fought a tiger.

Struggle: Lived on the run while building a web of rebels across provinces; the Raj put rewards on their heads.

Education: Worked as a clerk to hide movement; learned tactics, espionage and finance.

Achievement: Invented urban-style bank-robberies by using taxis, planned cross-province arms imports, led deadly jungle shootouts. He ambushed police, then melted away into the trees.

Impact: Made the revolution look like an organised crime syndicate the British couldn't control — fearless, smart, and brutal.



Inspiring Stories : Chittu Pandey

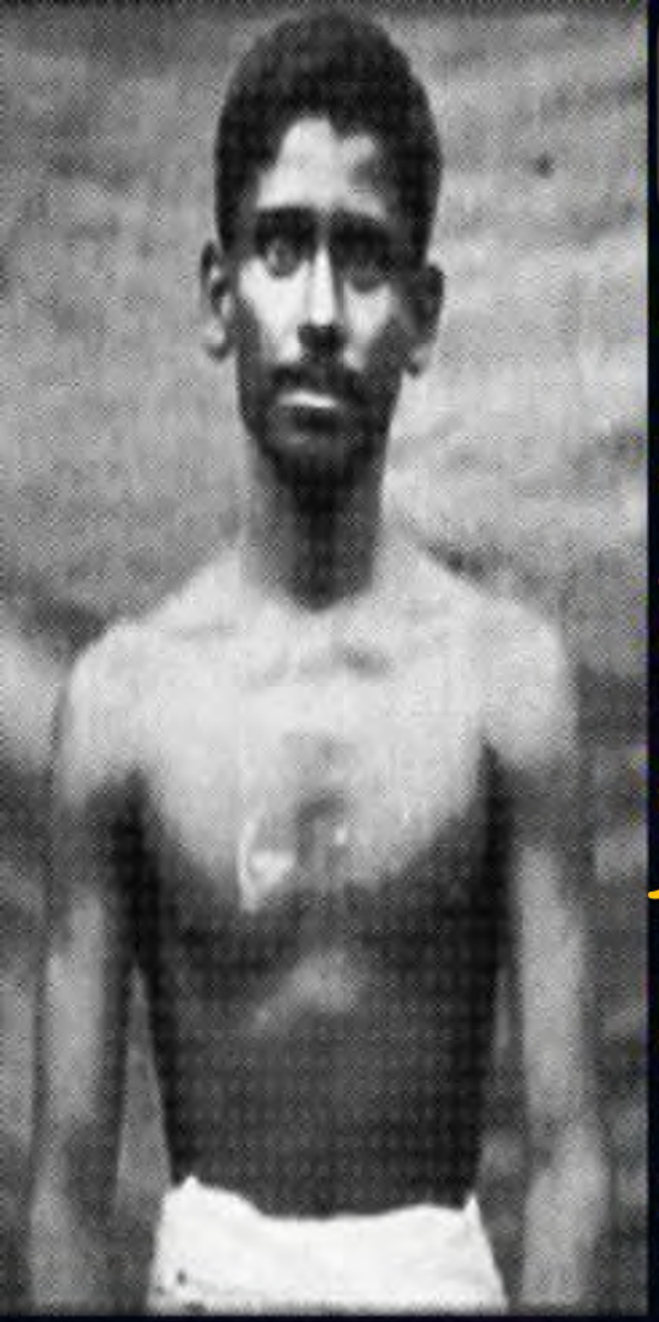
Chittu Pandey (Sher-e Ballia) Background: Small-town boss turned freedom chief in Ballia district.

Struggle: In 1942 the British tried mass arrests — normal leaders were jailed. Someone had to act fast.

Education: Grassroots leader who learned politics in the street, not the university.

Achievement: During Quit India he led a rapid local overthrow — forced the British collector to hand over power, declared a short-lived parallel government, and ran a small, roving strike force that acted like a rebel gang on the offensive.

Impact: For a few explosive days his crew ran the district — took the treasury, freed prisoners, and humiliated colonial rule. It was proof that small, fierce local bands could topple power, even if only briefly.



Inspiring Stories : Biren Datta Gupta

Biren (Birendranath) Datta Gupta Background: Young revolutionary tied to Bagha Jatin's circle. Quiet but deadly.

Struggle: The police were brutal, plotting and planting evidence; revolutionaries had to hit back or die.

Education: Learned underground tradecraft from older revolutionaries.

Achievement: Walked into the Calcutta High Court, pulled a revolver, and shot the top cop Shamsul Alam at point-blank range — then tried to escape like a one-man gangster movie.

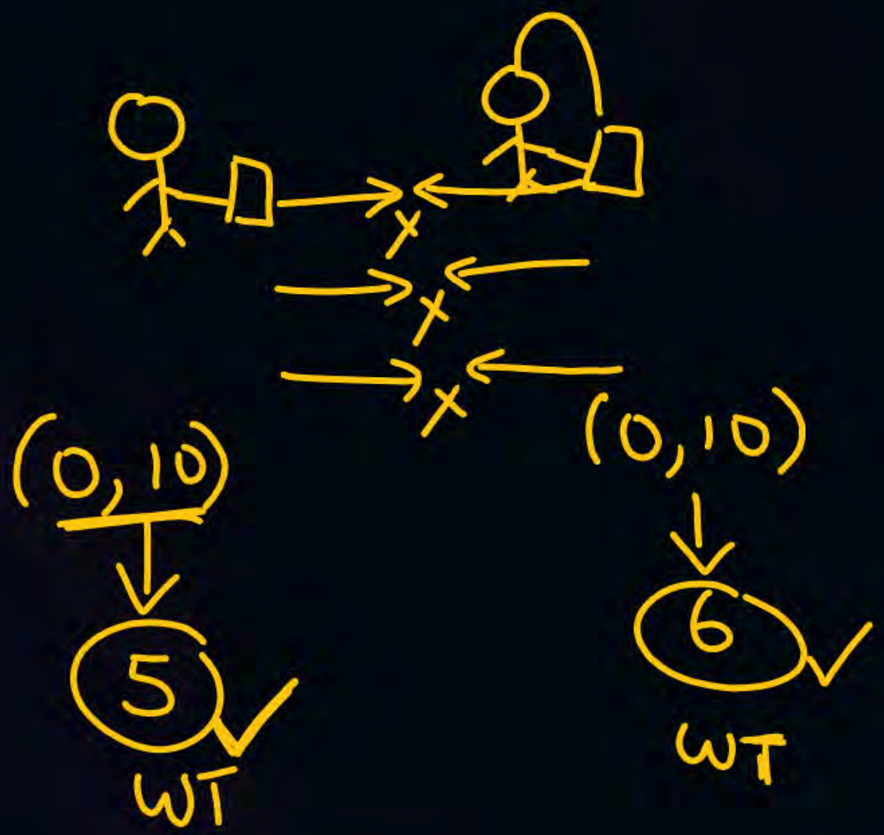
Impact: His courtroom assassination showed the British that revolutionaries could strike anywhere, even in the legal heart of the city. He was captured and later executed — died smiling, refusing to betray comrades.



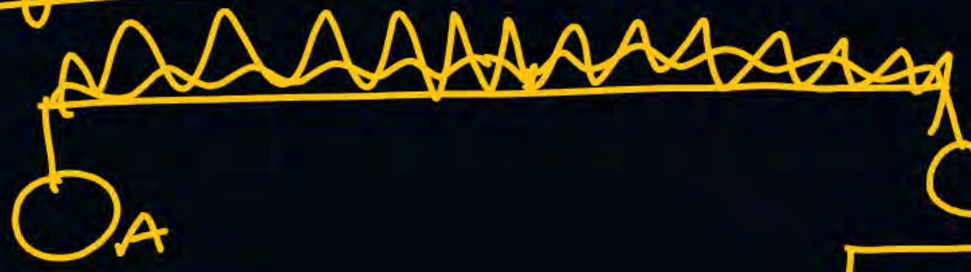
A has full control on the access link if the I bit of A reaches the other side.

monopolization can occur. To avoid monopolization, there is a max limit for a station to send data (1500B)

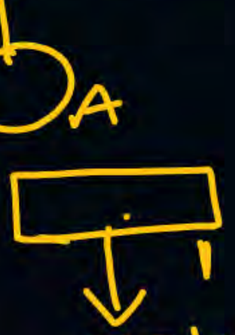

A
B
A
B
A
B


(0,10)
5
WT
(0,10)
6
WT

Binary Exponential Back off algorithm :-



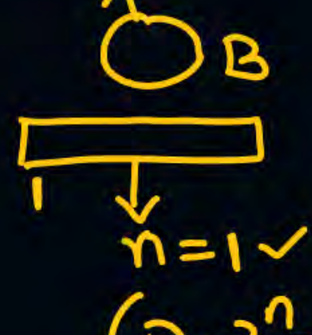
a) repeat
b) dont repeat



$(0, 2^n - 1)$

$(0, 2^1 - 1)$

$(0, 1)$



$(0, 2^n - 1)$

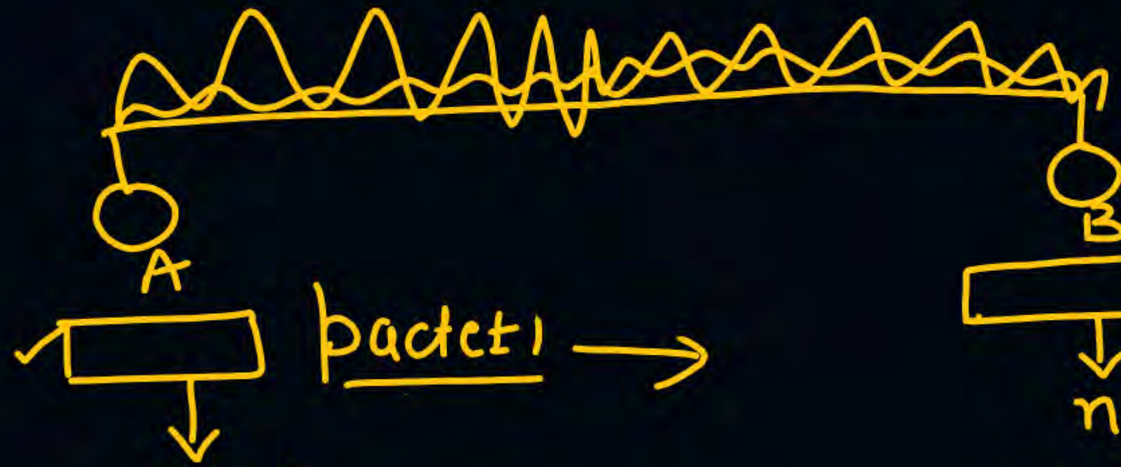
$(0, 1)$



$WT(A) = 0 * T_{slot}$ ✓ $= 0$

$WT(B) = 1 * T_{slot}$ ✓

$P(C) = \frac{2}{4}$ $P(A) = \frac{1}{4}$ $P(B) = \frac{1}{4}$



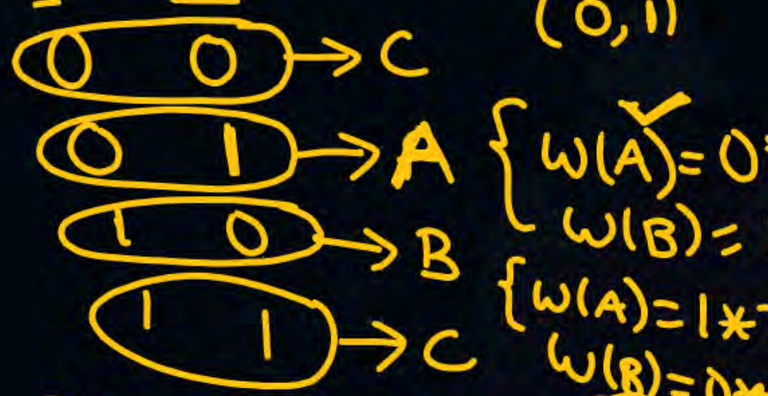
$(0, 2^n - 1)$

$(0, 2^1 - 1)$

$(0, 1)$

$(0, 2^n - 1)$

$(0, 1)$



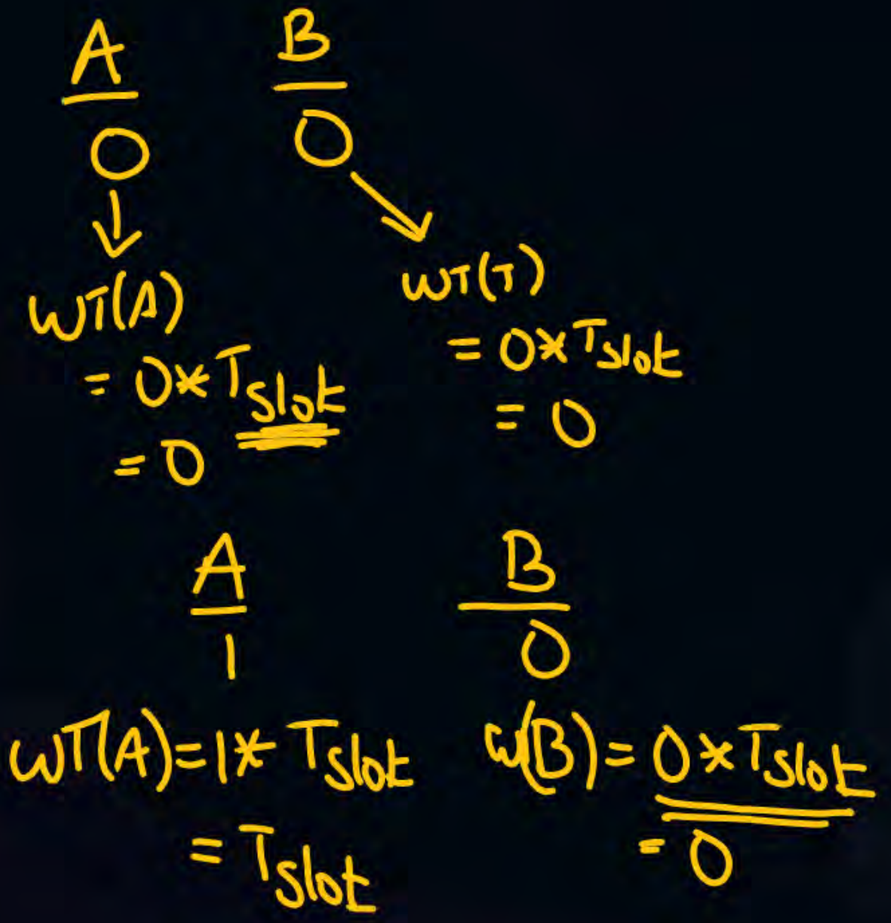
$\{W(A) = 0 * T_{slot}$
$W(B) = 1 * T_{slot}\}$

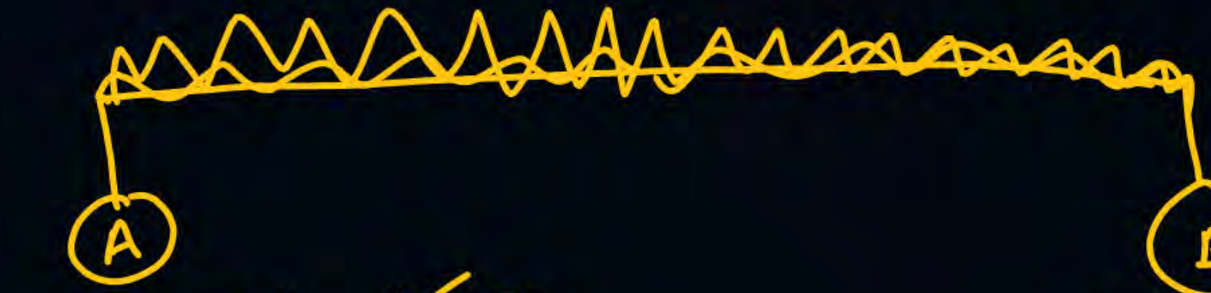
$\{W(A) = 1 * T_{slot}$
$W(B) = 0 * T_{slot}\}$

$P(C) = 2/4, \quad P(A) = 1/4 \quad P(B) = 1/4$

Assume:

A won the collision.


A
0
B
0
WT(A)
= 0 * T_Slot
= 0
WT(T)
= 0 * T_Slot
= 0
A
1
B
0
WT(A) = 1 * T_Slot
= T_Slot
W(B) = 0 * T_Slot
= 0



packet 2

Collision num = 1

$n = 1$ ✓

$(0, 2^n - 1)$

$(0, 1)$

Assume A won:-

packet 1

Collision 2

$n = 2$ ✓

$(0, 2^2 - 1)$

$(0, 1, 2, 3)$

A	B
0	0 – C
0	1 – A
0	2 – A
0	3 – A
1	0 – B
1	1 – C
1	2 – A
1	3 – A

$P(C) = 2/8$ ✓

$P(A) = 5/8$ ✓

$P(B) = 1/8$ ✓



A

B

packet = 3

Collision number = 1

$(0, 2^n - 1)$

$(0, 1)$

$WT = K * T_{slot}$

$K \in (0, 2^{n-1})$

$n \rightarrow$ Collision number

A	B
0	0 → C
0	1 → A
0	2 → A
0	3 → A
0	4 → A
0	5 → A
0	6 → A
0	7 → A

A	B
1	0 → B
1	1 → C
1	2 → A
1	3 → A
1	4 → A
1	5 → A
1	6 → A
1	7 → A

P = 1

Collision = 3

$(0, 2^3 - 1)$

$(0, 7)$

$P(C) = 2/16$

$P(B) = 1/16$

$P(A) = 13/16$

Collision probability is decreasing exponentially.

If a station, wins the collision, then its chances of winning the next collision increases

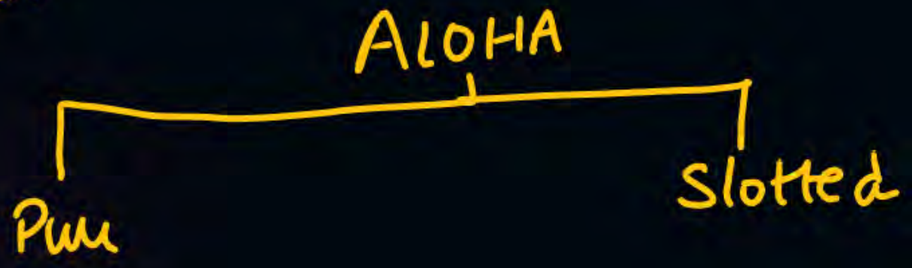
This is called capture effect

Back off algo is used only for 2 stations, therefore it is called binary backoff algo

ALOHA — Hello

→ Used in japan b/w the islands long back.
→ It is a wireless protocol
→ In this T_p → almost zero
→ Any station can transmit message anytime.
  no carrier sensing is required
→ Collisions are possible
  · No need to detect collisions as ack are present.

→ If a packet is lost in collision, then there will be retransmission, after some random amount of time called Backoff time.



Pure aloha :-

Vulnerable time = $2 \times T_t$



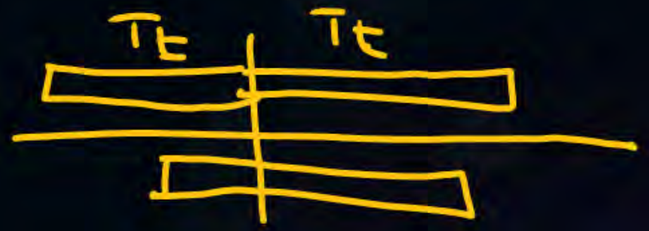
$\eta = G * e^{-2G}$ $\quad VT = 2 * Tt$

$G \rightarrow$ no of stations which wants to transmit in Tt.

$$\frac{d\eta}{dG} = 0 \Rightarrow G = \underline{1/2} \rightarrow \underline{\eta_{max}}$$

$1\,Tt = \frac{1}{2}$ Stations

$\underline{2\,Tt = 1\ \text{Station}}$



$$\eta = G * e^{-2G}$$

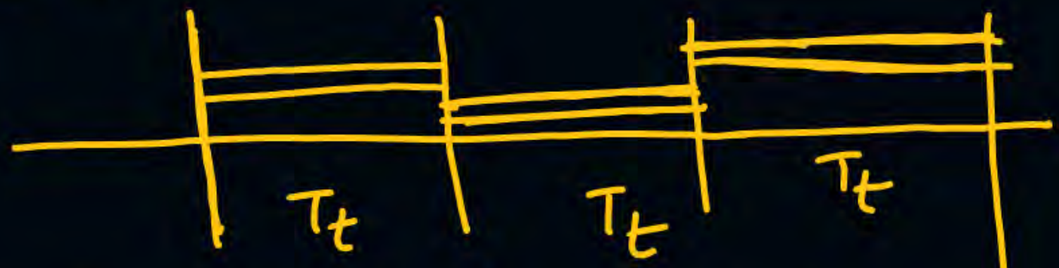
$G = 1/2 \rightarrow \eta_{max}$

$$\eta = 1/2 * e^{-2 * 1/2}$$

$$= 1/2\, e^{-1} = 0.184$$

η_{max} of pure ALOHA = 18.4%

Slotted aloha :-



$VT = 1 * T_t$

Time is divided into slots, each slot is 'T_t'

Stations should start transmission only at the beginning of the time slot



$$\eta = G * e^{-G}$$

$$\frac{d\eta}{dG} = 0 \Rightarrow \underline{\underline{G = 1}} \longrightarrow \underline{\underline{\eta_{max}}}$$

$G \rightarrow$ no of stations which wants to send data in T_t.

$$\eta_{max} = 1 \times e^{-1} = 0.368$$

η_{max} for slotted aloha = 36.8%

FCF

$$S\&W = \frac{1}{1+2a}$$

$$GBN, SR = \frac{N}{1+2a}$$

AC :-

$$TDM - \frac{1}{1+a}$$

$$Polling = \frac{T_t}{T_{Polling} + T_t + T_p}$$

$$CSMA/CD = \frac{1}{1+6.44a}$$

Aloha
- Pure: $G * e^{-2G}$ — $\eta_{max} = 18.4\%$
- Slotted: $G \times e^{-G}$ — $\eta_{max} = 36.8\%$

THANK - YOU